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Devices And Methods For Object Location Identification

Abstract

Embodiments relate to methods and systems for object location tracking of one object from among many objects in a same vicinity. An example method includes: affixing a location tracking device to each of the one object and the many objects, the location tracking device being configured to: detect a movement state of the location tracking device; in response to detecting the movement state, establish a connection to identify itself to an external server and to identify a location of the location tracking device and the one object; and if the connection is established, then allow connection with one or more other location tracking devices, so that the one or more other location tracking devices can be identified to the external server to identify the location of the one or more location tracking devices.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a Continuation of U.S. patent application Ser. No. 18/358,099, entitled “Devices and methods for object location identification”, filed Jul. 25, 2023, which claims priority to Australian patent application No. 2023901347, entitled “Devices and methods for object location identification”, filed May 4, 2023; the contents of which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to location tracking devices and methods. In particular, embodiments relate to devices and methods for identifying and tracking the location of an object.

BACKGROUND

[0003] Radio frequency (RF) waves are commonly used to move data wirelessly between devices. Like all technology, RF signals have some limitations. RF signals are broadcast and received at certain bands and channels within those bands. While being good at penetrating a lot of materials, RF signals cannot penetrate everything, and are very susceptible to attenuation when passing through glass, steel, and timber. An RF signal can be significantly weakened each time it comes into contact with one of these materials.

[0004] While an RF signal can be heavily attenuated when passing through certain materials, it is unusual for it be stopped completely. The RF signal gets weaker over distance, and the materials that are very difficult to penetrate shorten the readable distance of the RF signal considerably. This poses an issue in a warehouse environment, where these type of materials, especially steel and timber, are prevalent, such as in the form of pallets or steel cages. In particular, timber is especially good at absorbing an RF signal. This has the result that sending RF signals between RF devices in a warehouse with numerous wood or steel objects becomes unreliable and difficult.

[0005] It is desired to address or ameliorate one or more shortcomings or disadvantages of prior RF devices, or to at least provide a useful alternative thereto.

[0006] Throughout this specification the word “comprise”, or variations such as “comprises” or “comprising”, will be understood to imply the inclusion of a stated element, integer or step, or group of elements, integers or steps, but not the exclusion of any other element, integer or step, or group of elements, integers or steps.

[0007] Any discussion of documents, acts, materials, devices, articles or the like which has been included in the present specification is not to be taken as an admission that any or all of these matters form part of the prior art base or were common general knowledge in the field relevant to the present disclosure as it existed before the priority date of each of the appended claims.

SUMMARY

[0008] Some embodiments relate to a method of pallet location tracking of one pallet from among many pallets in a same vicinity, the method including: [0009] affixing a location tracking device to each of the one pallet and the many pallets, the location tracking device being configured to: [0010] detect a movement state of the location tracking device; [0011] in response to detecting the movement state, establish a connection to identify itself to an external server; and [0012] if the connection is established, then allow connection with one or more other location tracking devices, so that the one or more other location tracking devices can be identified to the external server.

[0013] The location tracking device may be further configured to, if the connection is not established, disable all functionality of the location tracking device not required to detect the movement state of the location tracking device.

[0014] The method may further include, after the affixing, receiving at the external server device identification data from the location tracking device of the one pallet, wherein the device identification data uniquely identifies the one pallet.

[0015] The method may further include, after the affixing, receiving at the external server device identification data from the location tracking device of the one pallet and the location tracking devices of the many pallets.

[0016] The device identification data from the location tracking devices of the many pallets may be received from the location tracking device of the one pallet.

[0017] The method may further include, after the receiving, accessing the external server to identify a location of the one pallet.

[0018] The movement state may be detected based on output of a movement sensor in each location tracking device, wherein detection of the movement state includes detecting when the location tracking device has moved for a first minimum time period followed by no movement for a second minimum time period. The first minimum time period may be between about 5 seconds to about 30 seconds. The second minimum time period may be between about 5 seconds to about 30 seconds.

[0019] Some embodiments relate to a system for pallet location tracking including: [0020] multiple location tracking devices, each location tracking device carried by a pallet, the location tracking device being configured to: [0021] detect a movement state of the location tracking device; [0022] in response to detecting the movement state, establish a connection to identify itself to an external server; and [0023] if the connection is established, then allow connection with one or more other location tracking devices, so that the one or more other location tracking devices can be identified to the external server; and [0024] the external server, wherein the external server is configured to identify a location of each pallet based on a location of the respective identified location tracking device carried by the pallet.

[0025] Each location tracking device may include a movement sensor and the movement state is detected based on output of the movement sensor, wherein detection of the movement state includes detecting when the location tracking device has moved for a first minimum time period followed by no movement for a second minimum time period. The first minimum time period may be between about 5 seconds to about 30 seconds. The second minimum time period may be between about 5 seconds to about 30 seconds.

[0026] Some embodiments relate to a location tracking device to be carried by an object for tracking the location of the object, wherein the location tracking device is a first location tracking device comprising: [0027] processing circuitry; [0028] memory accessible to the processing circuitry, wherein the memory is configured to store first device data of the first location tracking device, the first device data including location data; [0029] a movement sensor to detect movement of the first location tracking device; [0030] a low-power short-range radio frequency (RF) communications module; [0031] a battery to supply power to the processing circuitry, the memory, the movement sensor, and the communications module; [0032] a housing to house the processing circuitry, the memory, the movement sensor, the communications module, and the battery; and [0033] wherein the memory stores instructions that, when executed by the processing circuitry, cause the processing circuitry to: [0034] detect a movement state, based on output of the movement sensor, in which the first location tracking device has moved for a first minimum time period followed by no movement for a second minimum time period; [0035] in response to detecting the movement state, attempt to initiate wireless communication to a gateway device, via the RF communications module; [0036] if the attempt to initiate wireless communication to the gateway device is successful, then set an operational mode of the first location tracking device to a help

mode to allow communication with at least one third location tracking device, via the RF communications module; [0037] if the attempt to initiate wireless communication to the gateway device is unsuccessful, then attempt to initiate communication to a second location tracking device, via the RF communications module; [0038] if the attempt to initiate wireless communication to the second location tracking device is unsuccessful, then set an operational mode of the first location tracking device to a sleep mode; [0039] if the attempt to initiate wireless communication to the second location tracking device is successful, then set an operational mode of the first location tracking device to the help mode to allow communication to at least one third location tracking device, via the RF communications module; [0040] wherein in the help mode, the processing circuitry is configured to execute instructions to: [0041] allow communication with the at least one third location tracking device, via the RF communications module, for a third minimum time period; [0042] transmit the first device data and received device data from the at least one third location tracking device to the gateway device or the second location tracking device, via the RF communications module; and [0043] set the operational mode of the first location tracking device to the sleep mode after transmitting the first device data and received device data from the at least one third location tracking device.

[0044] Transmitting the first device data and received device data may further include transmitting, when no third location tracking devices are in communication with the first location tracking device, the first device data and received device data from the at least one third location tracking device to the gateway device or the second location tracking device, via the RF communications module.

[0045] The memory may store further instructions that, when executed by the processing circuitry, cause the processing circuitry to exit the sleep mode, if the first location tracking device is in the sleep mode when the movement state is detected. In the sleep mode, RF communication functions of the first location tracking device may be disabled. In the sleep mode, all functions of the first location tracking device not required in detecting the movement state may be disabled.

[0046] The first minimum time period may be between about 5 seconds and about 30 seconds. The second minimum time period may be between about 5 seconds and about 30 seconds. The help mode may be performed for a third minimum time period between about 1 second and about 30 seconds.

[0047] The movement sensor may have a power consumption of between about 30 nanoAmperes in a power-down mode to about 1 microAmpere in a normal operational mode.

[0048] The first location tracking device may be configured to attempt to initiate wireless communication with the gateway device for a first maximum time period. The first location tracking device may be configured to attempt to initiate wireless communication with the second location tracking device for a second maximum time period after the attempt to communicate with the gateway device is unsuccessful.

[0049] The first location tracking device may further comprise: [0050] a WiFi communications module and a mobile data communications module powered by the battery and housed in the housing; and [0051] wherein the memory further stores instructions that, when executed by the processing circuitry, cause the processing circuitry to: [0052] if the attempt to initiate wireless communication to the second location tracking device via the RF communications module is unsuccessful, then perform a scan to identify at least one WiFi access point, via the WiFi communications module; [0053] if the scan does not identify at least one WiFi access point, then set an operational mode of the first location tracking device to the sleep mode; and [0054] if the scan identifies at least one WiFi access point, then: [0055] receive, via the WiFi communications module, WiFi access point data from the WiFi access point; [0056] transmit, via the mobile data communications module, the WiFi access point data to an external server for identifying a location of the first location tracking device based on the transmitted WiFi access point data; and [0057] set an operational mode of the first location tracking device to the help mode to allow communication,

via the RF communications module, to at least one third location tracking device.

[0058] In the help mode, transmitting the first device data and received device data may include:

[0059] transmitting the first device data and the received device data from the at least one third location tracking device to: [0060] the gateway device, if communication to the gateway device via the RF communications module is successful; [0061] the second location tracking device, if communication to the second location tracking device via the RF communications module is successful; or [0062] to the external server via the mobile data communications module.

[0063] In the help mode, the instructions may further include instructions to: [0064] prior to transmitting the first device data and the received device data from the at least one third location tracking device to the external server via the mobile data communications module, attempt to initiate wireless communication to at least one of: another gateway device or another second location tracking device that includes a mobile data communications module.

[0065] In the help mode, the processing circuitry may be configured to execute instructions to: [0066] allow communication from the at least one third location tracking device, via the RF communications module, for a third minimum time period; [0067] transmit the first device data and the received device data from the at least one third location tracking device to: [0068] the gateway device, if communication to the gateway device via the RF communications module is successful; [0069] the second location tracking device, if communication to the second location tracking device via the RF communications module is successful; or [0070] attempt to initiate wireless communication with at least one of another gateway device or another second location tracking device that includes a mobile data communications module; [0071] if the attempt to initiate wireless communication to the another gateway device or another second location tracking device that includes a mobile data communications module is successful, transmit the first device data and the received device data from the at least one third location tracking device to the another gateway device or the another second location tracking device and set the operational mode of the first location tracking device to the sleep mode; [0072] if the attempt to initiate wireless communication to the another gateway device or another second location tracking device that includes a mobile data communications module is unsuccessful, transmit the first device data and received device data to the external server via the mobile data communications module and set the operational mode of the first location tracking device to the sleep mode.

[0073] The first location tracking device may be configured to attempt to initiate wireless communication with the at least one WiFi access point for a third maximum time period.

[0074] The first location tracking device may further comprise: [0075] a GNSS module powered by the battery and housed in the housing; and [0076] wherein the memory further stores instructions that, when executed by the processing circuitry, cause the processing circuitry to: [0077] if the attempt to initiate wireless communication to the at least one WiFi device, via the WiFi communications module, is unsuccessful, then attempt to determine a location of the first location tracking device via the GNSS module; [0078] if the attempt to determine the location of the first location tracking device via the GNSS module is successful, then: [0079] receive, via the GNSS module, GNSS data from a satellite constellation; [0080] transmit, via the mobile data communications module, GNSS data to an external server for identifying a location of the first location tracking device based on the transmitted GNSS data; and [0081] set an operational mode of the first location tracking device to the help mode to allow communication to at least one third location tracking device via the RF communications module;

[0082] The memory may further store instructions that, when executed by the processing circuitry, cause the processing circuitry to, if the attempt to determine the location of the first location tracking device via the GNSS module is unsuccessful, attempt to initiate wireless communication with the gateway device or the second location tracking device via the RF communications module until successful.

[0083] The first location tracking device may have a maximum dimension of about 50 mm and/or a

maximum mass of about 100 grams. For example, the mass may be about 20 grams to about 100 grams.

[0084] The first location tracking device may have a maximum dimension of about 200 mm and/or a maximum mass of about 100 grams. For example, the mass may be about 20 grams to about 100 grams.

[0085] The first location tracking device may have a maximum dimension of about 200 mm and/or a maximum mass of about 300 grams. For example, the mass may be about 100 grams to about 300 grams.

[0086] Some embodiments relate to a system including the first location tracking device of any one of the embodiments described above and another first location tracking device of any one of the embodiments described above, wherein the first location tracking devices have communicated with each other.

[0087] The system may further include multiple ones of the first location tracking device of any one of the embodiments described above, wherein each of the first location tracking devices has communicated with at least another one of the first location tracking devices.

[0088] In such systems, the first location tracking devices may be configured to cooperate to establish a network of wirelessly connected first location tracking devices.

[0089] The system may further include an external server, wherein at least one of the first location tracking devices is configured to communicate with the external server.

[0090] Some embodiments relate to a kit comprising at least two of the first location tracking device of the embodiments described above, wherein the at least two first location tracking devices are configured to communicate with each other.

[0091] Some embodiments relate to a method of object location tracking including configuring multiple types of the first location tracking devices of any one of the embodiments described above to be wirelessly connectable with each other.

[0092] The method may further comprise locating each of the first location tracking devices on a respective object to be tracked. The method may further comprise receiving at a server device data from at least one of the first location tracking devices to enable location identification of the first location tracking devices.

[0093] Some embodiments relate to use of multiple ones of the first location tracking devices of any one of the embodiments described above to establish a network of wirelessly connected first location tracking devices to track the location of multiple of objects carrying respective first location tracking devices.

[0094] Some embodiments relate to a pallet carrying the first location tracking device of any one of the embodiments described above.

[0095] Some embodiments relate to a method of object location tracking of one object from among many objects in a same vicinity, the method including: [0096] affixing a location tracking device to each of the one object and the many objects, the location tracking device being configured to:

[0097] detect a movement state of the location tracking device; [0098] in response to detecting the movement state, establish a connection to identify itself to an external server; and [0099] if the connection is established, then allow connection with one or more other location tracking devices, so that the one or more other location tracking devices can be identified to the external server.

[0100] The location tracking device may be further configured to, if the connection is not established, disable all functionality of the location tracking device not required to detect the movement state of the location tracking device.

[0101] The method may further include, after the affixing, receiving at the external server device identification data from the location tracking device of the one object, wherein the device identification data uniquely identifies the one object.

[0102] The method may further include, after the affixing, receiving at the external server device identification data from the location tracking device of the one object and the location tracking

devices of the many objects.

[0103] The device identification data from the location tracking devices of the many objects may be received from the location tracking device of the one object. The method may further include, after the receiving, accessing the external server to identify a location of the one object.

[0104] In such methods, the movement state may be detected based on output of a movement sensor in each location tracking device, wherein detection of the movement state includes detecting when the location tracking device has moved for a first minimum time period followed by no movement for a second minimum time period. The first minimum time period may be between about 5 seconds to about 30 seconds. The second minimum time period may be between about 5 seconds to about 30 seconds.

[0105] Some embodiments relate to a system for object location tracking including: [0106] multiple location tracking devices, each location tracking device carried by an object, the location tracking device being configured to: [0107] detect a movement state of the location tracking device; [0108] in response to detecting the movement state, establish a connection to identify itself to an external server; and [0109] if the connection is established, then allow connection with one or more other location tracking devices, so that the one or more other location tracking devices can be identified to the external server; and [0110] the external server, wherein the external server is configured to identify a location of each object based on a location of the respective identified location tracking device carried by the object.

Each location tracking device may include a movement sensor and the movement state may be detected based on output of the movement sensor. Detection of the movement state may include detecting when the location tracking device has moved for a first minimum time period followed by no movement for a second minimum time period. The first minimum time period may be between about 5 seconds to about 30 seconds. The second minimum time period may be between about 5 seconds to about 30 seconds.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0111] Embodiments of the present disclosure will now be described by way of non-limiting example only with reference to the accompanying drawings, in which:

[0112] FIG. 1 is schematic illustration of a system of wirelessly connected location tracking devices, according to some embodiments;

[0113] FIG. 2 is a schematic illustration of a location tracking device of FIG. 1, according to some embodiments;

[0114] FIG. 3 is a schematic illustration of a location tracking device of FIG. 1, according to some further embodiments;

[0115] FIG. 4 is a schematic illustration of a location tracking device of FIG. 1, according to still further embodiments;

[0116] FIG. 5 is a process flow diagram of a method of determining the location of the location tracking device of FIG. 2, according to some embodiments;

[0117] FIG. 6 is a process flow diagram of a method of determining the location of the location tracking device of FIG. 3, according to some embodiments;

[0118] FIG. 7 is a process flow diagram of a method of determining the location of the location tracking device of FIG. 4, according to some embodiments; and

[0119] FIG. 8 is a process flow diagram of a method of object location tracking of one object from among many objects in a same vicinity, according to some embodiments.

DESCRIPTION OF EMBODIMENTS

[0120] Described embodiments of the present disclosure relate to location tracking devices and

methods. In particular, embodiments relate to devices and methods for identifying and tracking the location of an object.

[0121] Tracking of individual objects, such as wooden pallets, amongst a plurality of objects, such as a stack of pallets, using individual devices carried by the objects can be extremely difficult. For example, the readable range of radio frequency signals of the devices carried by the pallets can be drastically reduced due to difficulty of the signals in penetrating the wooden pallets. Signals from devices amongst the stack of pallets may not be readable from a device external to the stack of pallets, but signals from devices on the outer edges of the stack of pallets may be readable. Further, devices amongst the stack of pallets may only be able to transmit a readable signal through one or two pallets in the stack before the signal becomes unreadable.

[0122] The following described embodiments relate to devices and methods for forming a mesh network of location tracking devices that can communicate with one another to allow devices amongst a plurality of objects, such as within the stack of pallets, that cannot transmit readable signals externally of the plurality of objects to instead transmit readable signals to other devices amongst the plurality of objects that can transmit readable signals externally. This allows devices that would not otherwise be able to transmit signals externally of the plurality of objects to transmit said signals externally via other devices amongst the plurality of objects.

[0123] Referring to FIG. 1, there is shown a system **10** of wirelessly connected location tracking devices **100**, **200**, **300**, according to some embodiments. The system **10** comprises a plurality of objects, such as objects **99**. The plurality of objects may be any one of a pallet, a trolley, a crate, or another object suitable for carrying goods, for example. For example, objects suitable for carrying goods to which an object location tracking device **100**, **200**, **300** may be coupled, affixed or otherwise located include: metal cages, shipping tanks, totes, crates or containers, metal and plastic material transport units, and durable plastic transport crates or tubs, such as intermediate bulk containers (IBCs). In some embodiments, the plurality of objects **99** are manufactured, at least in part, from any one of: steel, timber, and plastic. The objects **99** may or may not be the same kind of object.

[0124] Each object **99** carries a location tracking device **100**, **200**, **300** configured to allow identification of a location of the object **99**. Each of the location tracking devices **100**, **200**, **300** are configured to communicate with other location tracking devices **100**, **200**, **300** in its vicinity using wireless communications. The wireless connected location tracking devices **100**, **200**, **300** are configured to form a mesh network of wirelessly connected location tracking devices **100**, **200**, **300**. The mesh network of wirelessly connected location tracking devices **100**, **200**, **300** may allow for a location tracking device **100**, **200**, **300** to communicate with another location tracking device **100**, **200**, **300** via at least one another location tracking device **100**, **200**, **300** as an intermediary. That is, the location tracking devices **100**, **200**, **300** may communicate indirectly with other location tracking devices **100**, **200**, **300** via at least one another location tracking device **100**, **200**, **300**, for example.

[0125] In some embodiments, location tracking device **200** includes location tracking device **100** and may further includes WiFi communication and mobile data communication functionalities. That is, location tracking device **200** is location tracking device **100** with WiFi communication and mobile data communication capabilities, for example. However, in alternative embodiments, location tracking device **200** does not require WiFi communication functionality, but only requires the ability to see which WiFi hubs or other WiFi routing devices (i.e. access points) are in the vicinity of the location tracking device **200** without establishing any communications link with such WiFi devices. This allows the location tracking device **200** to gather and send information that can be used by an external server to determine a location of the location tracking device **200**. In some embodiments, location tracking device **300** includes location tracking device **200** and further includes global navigation satellite system communication functionality. That is, location tracking device **300** is location tracking device **200** with global navigation satellite system communication

capability, for example.

[0126] In some embodiments, system **10** further comprises a gateway device **130**. The gateway device **130** is a device that acts as a “gate” between the mesh network of wirelessly connected location tracking devices **100, 200, 300** and a network **150**. Each of the location tracking devices **100, 200, 300** are further configured to communicate with the gateway device **130** using RF communications. The location tracking devices **100, 200, 300** are configured to transmit device data to the gateway device **130**. The device data may include device identification data and configuration data (e.g. in the form of a configuration identifier) of the respective location tracking device **100, 200, 300**, for example. For versions of location tracking device **100, 200, 300** that have a GPS module to determine their own location, the device data may include location data that specifies the current location of that device. Otherwise, a location of the location tracking device **100, 200, 300** is determined by an external server based on the location of other devices in the vicinity, such as a gateway device or a WiFi device. For example, a gateway device has a known location, so any device that can locally communicate with the gateway device via RF communications is assumed to be at or close to the location of the gateway device. If multiple gateway devices are accessible to a location tracking device **100, 200, 300**, then the location of the location tracking device **100, 200, 300** is taken to be the location of the gateway that it is closest to, for example according to received signal strength.

[0127] In some embodiments, location data includes WiFi AP (access point) data. WiFi AP data may be publically available information relating to a WiFi access point, for example. In some embodiments, location data further includes global navigation satellite system (GNSS) data and assisted GNSS (AGNSS) data. Device identification data may include at least a device identification value of the respective location tracking device **100, 200, 300**. That is, each location tracking device **100, 200, 300** stores device identification data including a unique identifier allowing the respective location tracking device **100, 200, 300** to be uniquely identified, for example.

[0128] In some embodiments, configuration data includes a configuration identifier (ID) and optionally also a stocktake schedule or period. The configuration identifier can be used by the external server to determine a current configuration set of the location tracking device **100, 200, 300**, which allows the external server to determine whether a configuration update is required for that device. The configuration set corresponding to the configuration ID of the location tracking device **100, 200, 300** may specify, for example: how long the location tracking device **100, 200, 300** should scan for a gateway, once the movement state is detected; and how long to wait for movement and non-movement before determining that the movement state has been detected and that the device processing circuitry should “wake” up and try to identify itself.

[0129] The stocktake schedule or period may be separate from the configuration identifier. The stocktake schedule or period indicates at least one time or time period at which the location tracking device **100, 200, 300** should attempt to identify itself to an external server. In some embodiments, all location tracking devices **100, 200, 300** are configured to have the same stocktake schedule or period, so that all of the location tracking devices **100, 200, 300** attempt to identify themselves to the external server at or about the same time. For example, the location tracking devices **100, 200, 300** may be configured with a stocktake schedule time of 2300 hours GMT (Greenwich Mean Time) each day. This allows the location tracking devices **100, 200, 300** to effectively establish a network among themselves as needed, so that devices that would otherwise not be able to identify themselves from among a large number of devices or objects (e.g. due to noise and/or signal attenuation), can still get their device identification data out to the external server to have their locations identified.

[0130] The gateway device **130** may transmit device data to the location tracking devices **100, 200, 300**, such as configuration data, to update or overwrite existing device data, for example. The mesh network of wireless connected location tracking devices **100, 200, 300** allows location tracking

devices **100, 200, 300** unable to communicate directly with the gateway **130** to communicate indirectly with the gateway **130** via at least one another location tracking device **100, 200, 300**. That is, a location tracking device **100, 200, 300** may receive device data, such as configuration data, from the gateway device **130** via at least one another location tracking device **100, 200, 300**, for example.

[0131] Each (first) location tracking device **100, 200, 300** is configured to transmit device data to at least one other location tracking device **100, 200, 300** or the gateway device **130**, and to offer to help other location tracking devices **100, 200, 300**. If a communication connection is established with at least one other such location tracking device **100, 200, 300**, then the first location tracking device **100, 200, 300** can receive device data from at least one other location tracking device **100, 200, 300**. Each location tracking device **100, 200, 300** may receive data from up to a predetermined number (e.g. two to twenty) of other location tracking devices **100, 200, 300** simultaneously. That is, each location tracking device **100, 200, 300** may have up to twenty active connections simultaneously for transmitting and receiving data, for example. It is preferred to limit the predetermined number of active connections with other devices to a number less than 10, such as between 2 and 8, optionally around 5. This allows the device to conserve available energy in the battery, avoid exceeding its RAM (random access memory) limits and to limit the amount of message traffic in the area.

[0132] System **10** further comprises an external server **155**. External server **155** may be a physical server or a cloud-based server. External server **155** may comprise processing circuitry and a memory accessible to the processing circuitry. The processing circuitry may be configured to access data stored in memory, to execute instructions stored in memory **143**, and to read and write data to and from memory **143**. External server **155** is connected to the network **150**. That is, the external server **155** can be communicated with via network **150**, for example. The external server **155** is configured to receive and transmit data via network **150**.

[0133] In some embodiments, external server **155** is in communication with a database **157**. In some embodiments, the external server **155** includes the database **157**. The processing circuitry of the external server **155** may be configured to access data stored in database **157**, to execute instructions stored in database **157**, and to read and write data to and from database **157**.

[0134] The gateway device **130** is configured to communicate with the external server **155** via the network **150**. The gateway device **130** has a known physical location to the external server **155**. The gateway **130** may transmit device data received from the location tracking devices **100, 200, 300** to the external server **155**. The gateway **130** may receive device data from the external server to then transmit to the location tracking devices **100, 200, 300**. In some embodiments, a client **159** may communicate with the external server **155** via network **150**. The client **159** may be an owner of the plurality of objects, such as object **99**, and can track their owned objects via a web interface hosted on the external server **155**, for example.

[0135] Network **150** may comprise one or more local area networks or wide area networks that facilitate communication between elements of FIG. **1**. For example, according to some embodiments, network **150** may include the internet. However, network **150** may comprise at least a portion of any one or more networks having one or more nodes that transmit, receive, forward, generate, buffer, store, route, switch, process, or a combination thereof, etc. one or more messages, packets, signals, some combination thereof, or so forth. Network **150** may include, for example, one or more of: a wireless network, a wired network, an internet, an intranet, a public network, a packet-switched network, a circuit-switched network, an ad hoc network, an infrastructure network, a public-switched telephone network (PSTN), a cable network, a cellular network, a satellite network, a fibre-optic network, or some combination thereof.

[0136] In some embodiments, system **10** further comprises at least one WiFi access point (AP) **232**. Location tracking devices **200, 300** are further configured to wirelessly communicate with the at least one WiFi AP **232** via WiFi communications. Location tracking devices **200, 300** are further

configured to communicate with the external server **155** via mobile data communications. In some embodiments, the location tracking device **200, 300** communicates with at least one WiFi AP **232** to obtain WiFi AP data relating to the at least one WiFi AP **232**. In some embodiments, wireless communication with the at least one WiFi AP **232** comprises WiFi scanning to obtain WiFi AP data relating to the at least one WiFi AP **232**. That is, the location tracking devices **200, 300** do not perform a handshake with the at least one WiFi AP **232**, rather they obtain public information relating to the at least one WiFi AP **232** using WiFi scanning, for example.

[0137] The location tracking device **200, 300** then transmits the obtained WiFi AP data to the external server **155**, via mobile data communications. In some embodiments, the external server **155** provides the received WiFi AP data to a geolocation API, or a WiFi fingerprinting API, such as Google's Geolocation API, to determine a location of the location tracking device **200, 300**. That is, the external server **155** may determine the location of the location tracking device **200, 300** using WiFi fingerprinting, for example.

[0138] In some embodiments, system **10** further comprises a satellite constellation **312**. Location tracking devices **300** are further configured to wirelessly communicate with the satellite constellation **312** via GNSS communications. In some embodiments, the location tracking device **300** may communicate with the satellite constellation **312**, via GNSS communications, to obtain GNSS data. The GNSS data may be indicative of a geolocation of the location tracking device **300**. In some embodiments, location tracking devices **300** are further configured to wirelessly communicate with the external server **155**, via mobile data communications, to request AGNSS data from the external server **155**. In some embodiments, AGNSS data reduces the time to first fix (TTFF) of the location tracking device **300**. TTFF is a measure of the time required for the location tracking device **300** to acquire GNSS data from the satellite constellation **312**.

[0139] In some embodiments, the external server **155** may transmit device data to the location tracking devices **100, 200, 300**, such as configuration data, to overwrite existing device data, for example. The mesh network of wireless connected location tracking devices **100, 200, 300** allows location tracking devices **100, 200, 300** unable to communicate directly with the external server **155** to communicate indirectly with the external server **155** via at least one another location tracking device **100, 200, 300** that is in communication, either directly or indirectly, with a location tracking device **200, 300**. That is, a location tracking device **100, 200, 300** may receive device data, such as configuration data, from the external server **155** via at least one other location tracking device **100, 200, 300** that is in direct, or indirect, communication with the external server **155** via location tracking device **200, 300**, for example.

[0140] In some embodiments, system **10** is used to track the location of objects **99** using location tracking devices **100, 200, 300**. Objects **99**, being manufactured from signal-attenuating materials, such as steel or timber, may limit penetration of the wireless communication signals of the location tracking devices **100, 200, 300**, thereby reducing the readable distance of the signals. In situations where a location tracking device **100, 200, 300** is attempting to communicate with the external server **155**, but cannot do so directly, the location tracking device **100, 200, 300** may wireless communicate with another location tracking device **100, 200, 300** that is capable of initiating a wireless communication, either directly or indirectly, with the external server **155**. That is, location tracking devices **100, 200, 300** with wireless communication signals affected by objects **99**, causing reduced readable distance, may effectively (virtually) extend the readable distance of their signals using another location tracking device **100, 200, 300**, for example.

[0141] In some embodiments, it is beneficial for system **10** to include a combination of different types of location tracking devices **100, 200, 300**, so that the devices collectively include different functionality. In situations where the gateway **130** is not available at a location, location tracking devices **100** alone are unable to wirelessly communicate with the external server **155**, resulting in their location being untrackable. Providing location tracking device **200**, which the external server **155** can determine the location of using WiFi fingerprinting, at or near the location of the location

tracking devices **100**, enables the external server **155** to determine the location of the location tracking devices **100** via the location tracking device **200** if at least one WiFi AP **232** is available in the location.

[0142] In a situation where at least one WiFi AP **232** is unavailable, location tracking devices **100**, **200** alone are unable to wirelessly communicate with the external server **155**, resulting in their location being untrackable. Providing location tracking device **300**, which can determine its location using GNSS data and AGNSS data that can be subsequently transmitted to the external server **155**, at the location of the location tracking devices **100** enables the external server **155** to determine the location of the location tracking devices **100** via the location tracking device **300**. In some embodiments, the external server **155** is configured to receive device data from any one of the gateway device **130** and the location tracking devices **200**, **300**. The external server **155** may determine the location of the location tracking devices **100**, **200**, **300** based at least in part on the received device data.

[0143] For example, a gateway device **130** may receive from a first location tracking device **100**, **200**, **300** device data, the device data including at least device identification data, of the first location tracking device **100**, **200**, **300** and a plurality of location tracking devices **100**, **200**, **300** in either direct or indirect wireless communication with the first location tracking device **200**, **300**. Indirect wireless communication is communication of a location tracking device **100**, **200**, **300** of the plurality of location tracking devices **100**, **200**, **300** with the first location tracking device **100**, **200**, **300** via at least one another of the plurality of location tracking devices **100**, **200**, **300**. The gateway device **130** may then transmit the device data of the first location tracking device **100**, **200**, **300** and the plurality of location tracking devices **100**, **200**, **300** to the external server **155**. The external server **155**, knowing the location of the gateway device **130**, then assigns the first location tracking device **100**, **200**, **300** and each location tracking device **100**, **200**, **300** of the plurality of location tracking devices **100**, **200**, **300** indicated by the received device identification data, to the location of the gateway device **130**.

[0144] For example, the external server **155** may receive, from a first location tracking device **200**, **300** via mobile data communications, WiFi AP data retrieved by the first location tracking device **200**, **300** from at least one WiFi AP **232** at its respective location. The external server **155** may then receive from the first location tracking device **200**, **300** via mobile data communications, device data including device identification data, of the first location tracking device **200**, **300** and a plurality of location tracking devices **100**, **200**, **300** in either direct or indirect wireless communication with the first location tracking device **200**, **300**. Indirect wireless communication is communication of a location tracking device **100**, **200**, **300** of the plurality of location tracking devices **100**, **200**, **300** with the first location tracking device **200**, **300** via at least one another of the plurality of location tracking devices **100**, **200**, **300**. The external server **155** then determines the location of the first location tracking device **200**, **300** via WiFi fingerprinting, and assigns the first location tracking device **200**, **300** and each location tracking device **100**, **200**, **300** of the plurality of location tracking devices **100**, **200**, **300** indicated by the received device identification data, to the determined location of the first location tracking device **200**, **300**.

[0145] For example, the external server **155** may receive, from a first location tracking device **300** via mobile data communications, GNSS data retrieved by the first location tracking device **300** from a satellite constellation **312** indicating a location of the first location tracking device **300**. The external server **155** may then receive from the first location tracking device **300** via mobile data communications, device data including device identification data, of the first location tracking device **300** and a plurality of location tracking devices **100**, **200**, **300** in either direct or indirect wireless communication with the first location tracking device **300**. Indirect wireless communication is communication of a location tracking device **100**, **200**, **300** of the plurality of location tracking devices **100**, **200**, **300** with the first location tracking device **300** via at least one another of the plurality of location tracking devices **100**, **200**, **300**. The external server **155**, then

determines the location of the first location tracking device **300** using the received GNSS data, and assigns the first location tracking device **300** and each location tracking device **100**, **200**, **300** of the plurality of location tracking devices **100**, **200**, **300** indicated by the received device identification data, to the determined location of the first location tracking device **300**.

[0146] Identification of the location of the objects **99** may include the location tracking devices **100**, **200**, **300** identifying themselves to the external server **155**. The external server **155** may then determine the location of the location tracking devices **100**, **200**, **300**, and in turn their respective objects **99**, based on the method of identification, such as the gateway device **130**, via WiFi AP **232** scanning, or GNSS data. That is, the external server **155** may determine the location of location tracking devices **100**, **200**, **300** and their respective objects **99** based on how the location tracking devices **100**, **200**, **300** identify themselves with the external server **155**, for example.

[0147] In some embodiments, multiple ones of the location tracking device **100**, **200**, **300** are provided to establish system **10**, being a network of wirelessly connected location tracking devices **100**, **200**, **300** to track the location of multiple objects carrying respective location tracking devices **100**, **200**, **300**. The system **10** may include the external server **155** in combination with multiple ones of the location tracking devices **100**, **200**, **300**.

[0148] FIGS. **2**, **3**, and **4** are schematic illustrations of location tracking device **100**, location tracking device **200**, and location tracking device **300**, respectively, of FIG. **1**, according to various embodiments. Location tracking devices **100**, **200**, and **300** comprises processing circuitry **110** and a memory **112** accessible to the processing circuitry **110**. Processing circuitry **110** may be configured to access data stored in memory **112**, to execute instructions stored in memory **112**, and to read and write data to and from memory **112**. Processing circuitry **110** may comprise one or more microprocessors, microcontrollers, central processing units (CPUs), application specific instruction set processors (ASIPs), or other processor capable of reading and executing instruction code.

[0149] Memory **112** may comprise one or more volatile or non-volatile memory types, such as RAM, ROM, EEPROM, or flash, for example. Memory **112** may be configured to store executable applications or instruction code for execution by the processing circuitry **110**. Memory **112** may store device data, such as location data, device identification data, and configuration data of the location tracking device **100**.

[0150] Location tracking devices **100**, **200**, and **300** further comprise a movement sensor **114** accessible to the processing circuitry **110**. The movement sensor **114** is configured to monitor, detect, or measure the movement of the location tracking device **100**, **200**, **300**. In some embodiments, the movement sensor **114** is an accelerometer. The accelerometer may be a LIS2DW12 from STMicroelectronics, for example. In some embodiments, the movement sensor **114** is configured to draw minimal power when monitoring, detecting, or measuring. In some embodiments, the movement sensor **114** has a power consumption of between about 30 nanoAmperes (e.g. in a power-down mode) to about 1 microAmpere (e.g. in a normal but low power operational mode), for example. An example of a suitable movement sensor is the LIS2DW12 MEMS digital output motion sensor from STMicroelectronics, which is described as an ultra-low power consumption device. For example, the movement sensor may consume around 50 nA in power-down mode and about 1 microAmpere in low power mode.

[0151] Location tracking devices **100**, **200** and **300** further comprise a RF communications module **118** accessible to the processing circuitry **110**. The RF communications module **118** may facilitate communication of the location tracking device **100**, **200**, **300** with other location tracking devices **100**, **200**, **300** and/or the gateway device **130**. In some embodiments, the RF communications module **118** is configured to use a protocol, such as a Bluetooth protocol, Zigbee protocol, WiFi protocol or DECT (digital enhanced cordless telecommunications) for RF communication. In some embodiments, the RF communications module **118** is configured to use Bluetooth Low Energy (BLE) protocol for RF communications. In some embodiments, the processing circuitry **110**,

memory **112**, and the RF communications module **118** are included in a system on chip (SoC), such as a EFR32BG22 from Silicon Labs, for example.

[0152] Location tracking devices **200** and **300** further comprise a WiFi communications module **210** accessible to the processing circuitry **110**. The WiFi communications module **210** may facilitate communication of the location tracking device **200**, **300** with at least one WiFi AP **232**. In some embodiments, the WiFi communications module **210** is an SoC, such as a ESP8684 from Espressif Systems, for example.

[0153] Location tracking devices **200** and **300** further comprise a mobile data communications module **212**. The mobile data communications module **212** may facilitate communication of the location tracking device **200**, **300** with the external server **155**. In some embodiments, the mobile data communications module **212** is an LTE Cat NB2 module, such as a BC660K-GL from Quectel, for example. In other embodiments, the mobile data communications module **212** may be a CAT-3, CAT-2, CAT-M1 or CAT-NB1 module, for example.

[0154] Location tracking device **300** further comprises a GNSS module **310**. The GNSS module **310** facilitates communication of the location tracking device **300** with the satellite constellation **312**. The GNSS module **310** allows the location tracking device **300** to determine its geolocation. In some embodiments, the GNSS module **310** may be an LC76G from Quectel, for example.

[0155] Location tracking devices **100**, **200** and **300** further comprise a battery **116** for supplying power to the elements of the respective location tracking devices. Battery **116** of location tracking device **100** supplies power to the processing circuitry **110**, the memory **112**, the movement sensor **114** and the RF communications module **118**. Battery **116** of location tracking device **200** supplies power to the processing circuitry **110**, the memory **112**, the movement sensor **114**, the RF communications module **118**, the WiFi communications module **210**, and the mobile data communications module **212**. Battery **116** of location tracking device **200** supplies power to the processing circuitry **110**, the memory **112**, the movement sensor **114**, the RF communications module **118**, the WiFi communications module **210**, the mobile data communications module **212**, and the GNSS module **310**.

[0156] In some embodiments, the battery **116** of location tracking device **100** is a lithium battery, such as a CR2023 battery, for example. In some embodiments, location tracking device **100** may include a plurality of batteries **116**. That is, location tracking device **100** may include at least two CR2032 batteries, for example. The battery **116** of location tracking device **100** may have a capacity of at least 200 mAh (at 2 Volts), for example. The battery **116** of location tracking device **100** may have a capacity of 235 mAh (at 2 Volts), for example. In some embodiments, the battery **116** of location tracking device **100** is non-rechargeable. That is, the location tracking device **100** may not be a rechargeable device, for example.

[0157] In some embodiments, the battery **116** of location tracking device **200** is a lithium-ion battery. The battery **116** of location tracking device **200** may be a lithium-ion bag battery, for example. In some embodiments, location tracking device **200** may include a plurality of batteries **116**. That is, location tracking device **200** may include at least two lithium-ion batteries, for example. The battery **116** of location tracking device **200** may have a capacity of at least 1500 mAh, for example. The battery **116** of location tracking device **200** may have a capacity of 2000 mAh, for example. In some embodiments, the battery **116** of location tracking device **200** is non-rechargeable. That is, the location tracking device **200** may not be a rechargeable device, for example.

[0158] In some embodiments, the battery **116** of location tracking device **300** is a lithium-ion battery. The battery **116** of location tracking device **300** may be a lithium-ion bag battery, for example. In some embodiments, location tracking device **300** may include a plurality of batteries **116**. That is, location tracking device **300** may include at least two lithium-ion batteries, for example. The battery **116** of location tracking device **300** may have a capacity of at least 8000 mAh, for example. The battery **116** of location tracking device **300** may have a capacity of 10000

mAh, for example. In some embodiments, the battery **116** of location tracking device **300** is non-rechargeable. That is, the location tracking device **300** may not be a rechargeable device, for example.

[0159] In some embodiments, battery **116** may supply power to the processing circuitry **110** and memory **112** via a high efficiency step-up converter with ultra-low quiescent current, such as an ETA1061, for example. In some embodiments, battery **116** may supply power to the RF communications module **118** via a high efficiency step-up converter with ultra-low quiescent current, such as an ETA1061, for example. In some embodiments, battery **116** may supply power to the WiFi communications module **210** via a high efficiency step-up converter with ultra-low quiescent current, such as an ETA1061, for example. In some embodiments, battery **116** may supply power to the mobile data communications module **212** via a high efficiency step-up converter with ultra-low quiescent current, such as an ETA1061, for example. In some embodiments, battery **116** may supply power to the GNSS module **310** via a high efficiency step-up converter with ultra-low quiescent current, such as an ETA1061, for example.

[0160] Location tracking devices **100**, **200** and **300** further comprise housings **101**, **201**, and **301**, respectively, for housing the elements of the respective location tracking devices. Housing **101** of location tracking device **100** houses the processing circuitry **110**, the memory **112**, the movement sensor **114** and the RF communications module **118**. Housing **201** of location tracking device **200** houses the processing circuitry **110**, the memory **112**, the movement sensor **114**, the RF communications module **118**, the WiFi communications module **210**, and the mobile data communications module **212**. Housing **301** of location tracking device **300** houses the processing circuitry **110**, the memory **112**, the movement sensor **114**, the RF communications module **118**, the WiFi communications module **210**, the mobile data communications module **212**, and the GNSS module **310**. The location tracking device **100** is generally the smallest, while the location tracking device **300** is the largest and the location tracking device **200** is medium sized, in between the sizes of the devices **100** and **300**. However, the difference in sizes is primarily a result of the different componentry and functionality they carry, and so in some embodiments, the dimensions and/or mass of the different forms of the location tracking devices **100**, **200**, **300** may have some overlap or similarity. The location tracking device **100** may be somewhat circular, but in other embodiments may be roughly square, for example. The location tracking device **200** may be approximately square or rectangular or may be rounded, for example. The location tracking device **300** may be approximately rectangular, for example.

[0161] In some embodiments, the housings **101**, **201**, and **301** are manufactured from a suitable plastic material or other synthetic material. Examples of suitable plastics include polycarbonate and high density polyethylene (HDPE). The housing **101** of the location tracking device **100** may be sized to have a maximum dimension (e.g. length, width, depth or diameter) of around 50 mm. For example, a diameter or length of the housing **101** may be about 20 mm to about 50 mm. In some embodiments, the diameter or length of the housing **101** of the location tracking device **100** is about 24 mm. The housing **101** of the location tracking device **100** may have a width of about 20 mm to about 50 mm, for example. In some embodiments, the width of the housing **101** of the location tracking device **100** is about 24 mm. The housing **101** of the location tracking device **100** may have a height or thickness of about 10 mm to about 30 mm, for example. In some embodiments, the height or thickness of the housing **101** of the location tracking device **100** is about 12 mm. A mass of the location tracking device **100** may be up to about 100 grams. For example, the mass may be between about 20 grams and about 100 grams.

[0162] The housing **201** of the location tracking device **200** may be sized to have a maximum dimension (e.g. length, width, depth or diameter) of around 200 mm. For example, a length of the housing **201** may be about 25 mm to about 200 mm. In some embodiments, the length of the housing **201** of the location tracking device **200** is about 40 mm to 60 mm, optionally about 47 mm. The housing **201** of the location tracking device **200** may have a width of about 25 mm to

about 100 mm, for example. In some embodiments, the width of the housing **201** of the location tracking device **200** is about 40 mm to about 60 mm, optionally about 44 or 45 mm. The housing **201** of the location tracking device **200** may have a height or thickness of about 10 mm to about 30 mm, for example. In some embodiments, the height or thickness of the housing **201** of the location tracking device **200** is about 20 mm. A mass of the location tracking device **200** may be up to about 100 grams. For example, the mass may be between about 20 grams and about 100 grams.

[0163] The housing **301** of the location tracking device **300** may be sized to have a maximum dimension (e.g. length, width, depth or diameter) of around 200 mm. For example, a length of the housing **301** may be about 70 mm to about 200 mm. In some embodiments, the length of the housing **301** of the location tracking device **300** is about 140 mm. The housing **301** of the location tracking device **300** may have a width of about 40 mm to about 100 mm, for example. In some embodiments, the width of the housing **301** of the location tracking device **300** is about 80 mm. The housing **301** of the location tracking device **300** may have a height or thickness of about 10 mm to about 30 mm, for example. In some embodiments, the height or thickness of the housing **301** of the location tracking device **300** is about 20 mm. A mass of the location tracking device **300** may be up to about 300 grams. For example, the mass may be between about 100 grams and about 300 grams.

[0164] In some embodiments, location tracking device **100**, **200**, **300** is coupled to the object **99** via any one of: adhesion, mechanical fastening, and retention within a chamber of the object **99**, for example. In some embodiments, coupling of location tracking device **100**, **200**, **300** to the object **99** includes a combination of at least two of: adhesion, mechanical fastening, and retention within a chamber of the object **99**. Adhesion may include the use of at least one of: paste adhesive, liquid adhesive, film adhesive, and pellet adhesive, for example. Mechanical fastening may include the use of at least one of: screws, nails, nuts and bolts, zipties, rivets, and clamps, for example.

[0165] Referring to FIG. 5, there is shown a process flow diagram of a method **500** of determining the location of the location tracking device **100** of FIG. 2, hereinafter referred to as a “first location tracking device **100**”, according to some embodiments. At step **502** of method **500**, processing circuitry within the movement sensor **114** executes instructions, stored in its local memory, to cause an accelerometer component of the movement sensor **114** to monitor the movement of the first location tracking device **100**. This monitoring is performed while the movement sensor **114** is in a power-down mode that uses very little of its battery power.

[0166] At step **504**, the processing circuitry of the movement sensor **114** detects, based on the output of the accelerometer, a movement state of the first location tracking device **100**. Upon detecting the movement state, the movement sensor **114** generates and sends a signal to the processing circuitry **110** of the first location tracking device **100** to indicate that the movement state has been detected. In response to this detection of the movement state, the processing circuitry **110** causes the first location tracking device **100** to “wake up” from its sleep mode. In some embodiments, the movement state of the first location tracking device **100** comprises movement of the first location tracking device **100** for a first minimum time period immediately followed by no movement of the first location tracking device **100** for a second minimum time period. The movement state may also be described as a movement event, since it requires multiple conditions to be satisfied, namely a movement period for a certain time period, immediately followed by a non-movement period for another time period. If these conditions are not satisfied, then the movement state (or event) is not detected.

[0167] In some embodiments, the first minimum time period and the second minimum time period are the same. In some embodiments, the first minimum time period and the second minimum time period are different. In some embodiments, the first minimum time period is between about 5 seconds and about 30 seconds. In some embodiments, the first minimum time period is between about 8 seconds and about 15 seconds. In some embodiments, the first minimum time period is about 10 seconds. In some embodiments, the second minimum time period is between about 5

seconds and about 30 seconds. In some embodiments, the second minimum time period is between about 8 seconds and about 15 seconds. In some embodiments, the second minimum time period is about 10 seconds.

[0168] If the processing circuitry of the movement sensor **114** does not detect the movement state of the location tracking device **100**, method **500** proceeds back to step **502**. That is, steps **502** and **504** of method **500** will be repeated until the processing circuitry of the movement sensor **114** detects a movement state of the first location tracking device **100**, for example. If the processing circuitry of the movement sensor **114** detects the movement state of the first location tracking device **100**, method **500** proceeds to step **506**.

[0169] Steps **502** and **504** of method **500** may be performed in an operational mode referred to as a sleep mode. The sleep mode is an operational mode of the location tracking device **100** in which the location tracking device **100** preserves life of the battery **116** by minimising functions that are not required for detecting the movement state. That is, location tracking device **100** may disable functions not required for detecting the movement state to reduce power consumption and extend the life of the battery **116**, for example.

[0170] In some embodiments, in the sleep mode RF communication functions of the location tracking device **100** are disabled. In some embodiments, in the sleep mode, all functions of the location tracking device **100** not required in detecting the movement state are disabled. That is, any function of the location tracking device **100** not required to monitor movement of the location tracking device **100** via the movement sensor **114** and to detect the movement state is disabled, for example.

[0171] At step **506**, the processing circuitry **110** executes instructions, stored in memory **112**, to cause the first location tracking device **100** to attempt to initiate communication with the gateway device **130** via the RF communications module **118**. In some embodiments, the first location tracking device **100** attempts to initiate communication with the gateway device **130** using a low power short-range wireless radio frequency communication protocol, such as BLE. The first location tracking device **100** attempts to initiate communication with the gateway device **130** for a first maximum time period. The first maximum time period may be between about 200 milliseconds to about 2 seconds, optionally around 500 milliseconds, for example.

[0172] At step **508**, the processing circuitry determines whether initiating communication with the gateway device **130** was successful. If the connection is successful, method **500** proceeds to step **514**, to be described in detail below. If the connection is unsuccessful, method **500** proceeds to step **510**.

[0173] At step **510**, the processing circuitry **110** executes instructions, stored in memory **112**, to cause the first location tracking device **100** to attempt to initiate wireless communication with a second location tracking device **100**, **200**, **300** via the RF communications module **118**. The first location tracking device **100** attempts to initiate communication with the second location tracking device **100**, **200**, **300** for a second maximum time period. The second maximum time period may be between about 200 milliseconds to about 2 seconds, optionally about 500 milliseconds to about 1 second, for example.

[0174] The second location tracking device **100**, **200**, **300** may be a location tracking device **100**, **200**, **300** that has a direct communication channel, or an indirect communication channel via other location tracking device **100**, **200**, **300**, to the gateway device **130** which communicates with the external server **155**. The second location tracking device **100**, **200**, **300** may be a location tracking device **200**, **300** that has a direct communication channel to the external server **155** via the mobile data communications module **212**. The second location tracking device **100**, **200**, **300** may be a location tracking device **100**, **200**, **300** that has an indirect communication channel via other location tracking device **100**, **200**, **300**, to a location tracking device **200**, **300** that has a direct communication channel to the external server **155** via the mobile data communications module **212**.

[0175] In some embodiments, the first location tracking device **100** attempts to initiate wireless communication with the second location tracking device **100, 200, 300** using a low power short-range wireless radio frequency communication protocol, such as BLE. At step **512**, the processing circuitry **110** determines whether initiating communication with the second location tracking device **100, 200, 300** was successful. If the connection is unsuccessful, method **500** proceeds to step **520**, to be described in detail below. If the connection is successful, method **500** proceeds to step **514**.

[0176] At step **514**, the processing circuitry **110** executes instructions, stored in memory **112**, to set the operational mode of the first location tracking device **100** to a help mode. The first location tracking device **100** enters the help mode if a connection to either the gateway **130** or the second location tracking device **100, 200, 300** is established. The help mode is an operational mode of the location tracking device **100, 200, 300** in which the location tracking device **100, 200, 300** offers assistance to other location tracking devices **100, 200, 300** performing step **510**. That is, the help mode is a mode in which the location tracking device **100, 200, 300**, having a communication channel with the external server **155**, offers to extend that communication channel to other location tracking devices **100, 200, 300** attempting to transmit their device data to the external server **155**, for example.

[0177] After entering the help mode, method **500** proceeds to step **516**. At step **516**, the processing circuitry **110** executes instructions, stored in memory **112**, to cause the first location tracking device **100** to allow communication with one or more third location tracking devices **100, 200, 300** for a third minimum time period. In some embodiments, the third minimum time period is between about 1 second and about 30 seconds. In some embodiments, the third minimum time period is between about 5 seconds to about 20 seconds, optionally about 10 seconds. The processing circuitry **110** executes instructions, stored in memory **112**, to cause the first location tracking device **100** to allow for a further maximum time period, such as a period of between about 30 seconds to about 120 seconds, within which to communicate data from the first location tracking device **100** and data received from all connected third location tracking devices **100, 200, 300** to the gateway device **130**.

[0178] In some embodiments, the first location tracking device **100** allows simultaneous communication with a maximum number of third location tracking devices **100, 200, 300**. The maximum number of third location tracking devices **100, 200, 300** may be between 2 and 19, for example. In some embodiments, the third location tracking device **100, 200, 300** communicates with the first location tracking device **100** using a low power short-range wireless radio frequency communication protocol, such as BLE.

[0179] To allow communication with one or more third location tracking devices **100, 200, 300**, the processing circuitry executes instructions, stored in memory **112**, to cause the first location tracking device **100** to advertise that it is allowing wireless communication for a fourth time period, and to then scan for wireless communication requests for a fifth time period. In some embodiments the fourth time period is between about 10 ms and about 30 ms. In some embodiments, the fourth time period is about 20 ms. In some embodiments the fifth time period is between about 30 ms and about 50 ms. In some embodiments, the fifth time period is about 40 ms.

[0180] If no third location tracking devices **100, 200, 300** initiate communication with the first location tracking device **100** during the third minimum time period, method **500** proceeds to step **518**. If one or more third location tracking devices **100, 200, 300** initiate communication with the first location tracking device **100**, the first location tracking device **100** maintains the connection, waiting to receive device data from the connected one or more third location tracking devices **100, 200, 300**. The first location tracking device **100** maintains the connection for a first timeout period or until it receives the device data from the third location tracking device **100, 200, 300** (whichever happens first), after which the first location tracking device **100** terminates the connection. The first timeout period may be up to 60 seconds, for example, or in other examples it may be about 20, 10 or about 2 seconds. A second timeout period may apply from a time at which the first location

tracking device **100** enters the help mode. The second timeout period may be approximately double the first timeout period, for example. The second timeout period may be up to 100 or 120 seconds, for example, or in other examples it may be around 80, 60, 40, 30, 20, 10, 5 or 4 seconds. After the second timeout period expires, the first location tracking device **100** ceases to operate in the help mode, does not permit new connections with third location tracking devices **100, 200, 300** and terminates any existing connections, including any connections with any second location tracking devices **100, 200, 300**.

[0181] The received device data from the one or more third location tracking devices **100, 200, 300** comprises device data of the respective one or more third location tracking device **100, 200, 300**. In some embodiments, the received device data from the one or more third location tracking devices **100, 200, 300** further comprises device data of one or more other location tracking device **100, 200, 300**. That is, the one or more third location tracking devices **100, 200, 300** may have previously received device data from other location tracking devices **100, 200, 300** attempting to transmit their device data to the external server **155**, for example.

[0182] Upon receiving device data from a third location tracking device **100, 200, 300** of the one or more third location tracking device **100, 200, 300** in communication with the first location tracking device **100**, the third location tracking device **100, 200, 300** terminates the wireless connection with the first location tracking device **100**. The third location tracking device **100, 200, 300** having terminated the wireless connection, allows an additional third location tracking device **100, 200, 300** to initiate wireless communication with the first location tracking device **100**. In some embodiments, one or more third location tracking devices **100, 200, 300** may timeout or lose connection to the first location tracking device **100**. The one or more third location tracking devices **100, 200, 300** may attempt to reinitiate wireless communication with the first location tracking device **100** or another location tracking device **100, 200, 300**.

[0183] When the first location tracking device **100** no longer has any of the one or more third location tracking devices **100, 200, 300** connected via wireless communications, method **500** proceeds to step **518**. That is, when all of the one or more third location tracking device **100, 200, 300** have disconnected from the first location tracking device **100**, method **500** proceeds to step **518**, for example.

[0184] At step **518**, the processing circuitry **110** executes instructions, stored in memory **112**, to cause the first location tracking device **100** to transmit device data of the first location tracking device **100** and any received device data from the one or more third location tracking devices **100, 200, 300** to the connected wireless communication device, being the gateway device **130** or the second location tracking device **100, 200, 300**. In some embodiments, the first location tracking device **100** only transmits device data and received device data including identification data. After transmitting the device data and any received device data, the processing circuitry **110** of the first location tracking device **100** executes instructions, stored in memory **112**, to cause the RF communications module **118** of first location tracking device **100** to terminate the successfully initiated communication channel with the gateway device **130** or the second location tracking device **100, 200, 300**, and method **500** proceeds to step **520**.

[0185] At step **520**, the processing circuitry **110** executes instructions, stored in memory **112**, to set the operational mode of the first location tracking device **100** to the sleep mode.

[0186] Referring to FIG. **6**, there is shown a process flow diagram of a method **600** of determining the location of the location tracking device **200** of FIG. **3**, hereinafter referred to as a “first location tracking device **200**”, according to some embodiments. Steps **602** to **610** of method **600** of the first location tracking device **200** correspond to steps **502** to **510** of method **500** of the first location tracking device **100**, respectively, the description of which is not repeated here for simplicity. Location tracking devices **200** have all of the functionality of location tracking devices **100**, plus additional hardware and functionality to perform the method **600**, as described.

[0187] Steps **602** and **604** of method **600** may be referred to as a sleep mode. The sleep mode is an

operational mode of the location tracking device **200** in which the location tracking device **200** preserves life of the battery **116** by minimising functions that are not required for detecting the movement state. That is, location tracking device **100** may disable functions not required for detecting the movement state to reduce power consumption and extend the life of the battery **116**, for example.

[0188] In some embodiments, in the sleep mode RF communication functions of the location tracking device **200** are disabled. In some embodiments, in the sleep mode WiFi communications of the location tracking device **200** are disabled. In some embodiments, in the sleep mode mobile data communications of the location tracking device **200** are disabled. In some embodiments, in the sleep mode all functions of the location tracking device **200** not required in detecting the movement state are disabled. That is, any function of the location tracking device **200** not required to monitor movement of the location tracking device **200** via the movement sensor **114** and to detect the movement state are disabled, for example.

[0189] At step **612**, the processing circuitry **110** determines whether initiating communication with the second location tracking device **100**, **200**, **300** was successful. If the connection is successful, method **600** proceeds to step **620**, to be described in detail below. If the connection is unsuccessful, method **600** proceeds to step **614**.

[0190] At step **614**, the processing circuitry **110** executes instructions, stored in memory **112**, to cause the first location tracking device **200** to perform a scan to identify at least one WiFi access point **232** via the WiFi communications module **210**. At step **616**, the processing circuitry **110** determines whether the scan identified at least one WiFi AP **232**. If the scan did not identify at least one WiFi AP **232**, method **600** proceeds to step **626**, to be described in detail below. If the scan identifies at least one WiFi AP **232**, method **600** proceeds to step **618**.

[0191] At step **618**, the processing circuitry **110** executes instructions, stored in memory **112**, to cause the first location tracking device **200** to retrieve WiFi AP data from each connected WiFi AP **232** of the at least one WiFi APs **232**. In some embodiments, the WiFi AP data of each WiFi AP **232** comprises a WiFi access point objects array. The WiFi AP data of each WiFi AP **232** includes at least a MAC address of the respective WiFi AP **232** and the signal strength of the communication with the WiFi AP **232** measured in dBm, for example. Each respective WiFi access point objects array may include at least the MAC address of the respective WiFi AP **232** and the signal strength of the communication with the WiFi AP **232** measured in dBm, and optionally also at least one of: the number of milliseconds since WiFi AP **232** was detected, the channel over which the first location tracking device **200** and the WiFi AP **232** are communicating, and the signal to noise ratio measured in dB. The signal strength of the communication with the WiFi AP **232** allows the external server to determine an approximate distance of the first location tracking device **100**, **200**, **300** from the location of the WiFi AP **232**, which can also allow for location determination by triangulation where data is received about multiple such WiFi APs **232**.

[0192] Upon retrieving the WiFi AP data from each connected WiFi AP **232** of the at least one WiFi APs **232**, the processing circuitry **110** executes instructions, stored in the memory **112**, to cause the first location tracking device **200** to wirelessly transmit the retrieved WiFi AP data to the external server **155** via the mobile data communications module **212**. At step **619**, the external server **155** processes the transmitted WiFi AP data using a WiFi fingerprinting API, such as Google's Geolocation API, to determine the location of the first location tracking device **200**.

[0193] If the external server **155** determines the location of the first location tracking device **200**, the first location tracking device **200** then receives, from the external server **155**, an acknowledgement that location determination of the first location tracking device **200** was successful. If the external server **155** cannot determine the location of the first location tracking device **200**, the first location tracking device **200** then receives, from the external server **155**, an acknowledgement that location determination of the location tracking device **200** was unsuccessful. In some embodiments, the external server **155** may be configured to determine that the location

determination of the first location tracking device **200** is unsuccessful if an accuracy returned by the WiFi fingerprinting API is below a particular threshold, wherein the accuracy is indicative of the accuracy of the determined location.

[0194] Upon receiving an acknowledgement indicating an unsuccessful location determination of the location tracking device **200** from the external server **155**, method **600** proceeds to step **626**. Upon receiving an acknowledgement indicating a successful location determination of the location tracking device **200** from the external server **155**, method **600** proceeds to step **620**.

[0195] Steps **620** to **626** of method **600** of the first location tracking device **200** correspond to steps **514** to **520** of method **500** of the first location tracking device **100**, respectively.

[0196] Referring to FIG. 7, there is shown a process flow diagram of a method **700** of determining the location of the location tracking device **300** of FIG. 4, hereinafter referred to as a “first location tracking device **300**”, according to some embodiments. Steps **702** to **718** of method **700** of the first location tracking device **300** correspond to steps **602** to **618** of method **600** of the first location tracking device **200**, respectively, the description of which is not repeated here for simplicity.

Location tracking devices **300** have all of the functionality of location tracking devices **100**, **200**, plus additional hardware and functionality to perform the method **700**, as described.

[0197] Steps **702** and **704** of method **600** may be referred to as a sleep mode. The sleep mode is an operational mode of the location tracking device **300** in which the location tracking device **300** preserves life of the battery **116** by minimising functions that are not required for detecting the movement state. That is, location tracking device **100** may disable functions not required for detecting the movement state to reduce power consumption and extend the life of the battery **116**, for example.

[0198] In some embodiments, in the sleep mode RF communication functions of the location tracking device **300** are disabled. In some embodiments, in the sleep mode WiFi communications of the location tracking device **300** are disabled. In some embodiments, in the sleep mode mobile data communications of the location tracking device **300** are disabled. In some embodiments, in the sleep mode GNSS communications of the location tracking device **300** are disabled. In some embodiments, in the sleep mode all functions of the location tracking device **300** not required in detecting the movement state are disabled. That is, any function of the location tracking device **300** not required to monitor movement of the location tracking device **300** via the movement sensor **114** and to detect the movement state are disabled, for example.

[0199] At step **716**, the processing circuitry **110** determines whether the scan identified at least one WiFi AP **232**. If the scan identifies at least one WiFi AP **232**, method **700** proceeds to step **718**, as previously described. If the scan did not identify at least one WiFi AP **232**, method **700** proceeds to step **720**.

[0200] At step **720**, the processing circuitry **110** executes instructions, stored in memory **112**, to check whether valid AGNSS data is stored in memory **112** of the first location tracking device **300**. If no valid AGNSS data is stored in memory **112**, the processing circuitry **110** executes instructions, stored in memory **112**, to cause the first location tracking device **300** to wirelessly communicate with the external server **155** to request AGNSS data for the current location. Upon receiving the AGNSS data from the external server **155**, the processing circuitry **110** executes instructions, stored in memory **112**, to write the AGNSS data to memory **112**. That is, if invalid AGNSS data, or no AGNSS data, is stored in the device data of memory **112**, the processing circuitry **110** writes the received valid AGNSS data to the memory **112**, for example.

[0201] If valid AGNSS data is stored in memory **112**, or when the received valid AGNSS data is stored in memory **112**, method **700** proceeds to step **722**. At step **722** the processing circuitry **110** executes instructions, stored in memory **112**, to cause the first location tracking device **300** to attempt to initiate wireless communication with satellite constellation **312** via the GNSS module **310**. At step **724**, the processing circuitry **110** determines whether initiating communication with the satellite constellation was successful. If the connection is successful, the first location tracking

device **300** will receive GNSS data from the satellite constellation **312**, indicating a location of the first location tracking device **300**. Upon receiving the GNSS data from the satellite constellation **312**, method **700** proceeds to step **726**.

[0202] At step **726**, the processing circuitry **110** executes instructions, stored in the memory **112**, to cause the first location tracking device **300** to wirelessly transmit the received GNSS data to the external server **155** via the mobile data communications module **212**. The first location tracking device **300** then receives, from the external server **155**, an acknowledgement that the GNSS data was received by the external server **155**. Upon receiving the acknowledgement from the external server **155**, method **700** proceeds to step **732**.

[0203] Steps **732** to **738** of method **700** of the first location tracking device **300** correspond to steps **620** to **626** of method **600** of the first location tracking device **200**, respectively.

[0204] At step **724**, if the connection to the satellite constellation **312** is unsuccessful, method **700** proceeds to step **728**. At step **728**, the processing circuitry **110** executes instructions, stored in the memory **112**, to cause the first location tracking device **300** to attempt to initialise wireless communications with the gateway device and the second location tracking device **100, 200, 300**. At step **730**, the processing circuitry **110** determines whether initiating communication with the gateway device **130** or the second location tracking device **100, 200, 300** was successful. If the connection is successful, method **700** proceeds to step **732**, as previously described. If the connection is unsuccessful, method **700** proceeds to back to step **728**, as previously described. That is, the first location tracking device **300** will attempt to initialise wireless communication with the gateway device **130** and the second location tracking device **100, 200, 300** until a successful connection is made, for example.

[0205] In some embodiments, location tracking device **100** is configured to periodically perform steps **506** to **520** of method **500** without the movement state of the location tracking device **100** being detected. That is, the location tracking device **100** may periodically attempt to identify itself to the external server without moving, for example. In some embodiments, location tracking device **200** is configured to periodically perform steps **606** to **626** of method **600** without the movement state of the location tracking device **200** being detected. That is, the location tracking device **200** may periodically attempt to identify itself to the external server without moving, for example. In some embodiments, location tracking device **300** is configured to periodically perform steps **706** to **732** of method **700** without the movement state of the location tracking device **300** being detected. That is, the location tracking device **300** may periodically attempt to identify itself to the external server without moving, for example.

[0206] In some embodiments, each of the location tracking devices **100, 200, and 300** are configured to periodically attempt to identify themselves to the external server **155** at the same time. Each location tracking device **100, 200, 300** may be configured with the same stocktake schedule time, such that they are configured to attempt to identify themselves to the external server **155** at the same time or during a same time period. That is, all location tracking devices **100, 200, and 300** may perform the respective steps of methods **500, 600, and 700**, respectively, periodically at the same time or during a same time period based on the configured function set (e.g. corresponding to its configuration data) of the respective location tracking device **100, 200, 300** and data stored therein, including the stocktake schedule, for example.

[0207] In some embodiments, a kit comprising at least two location tracking devices **100, 200, 300** configured to communicate with one another is provided. The at least two location tracking devices **100, 200, 300** are configured to wirelessly communicate with one another.

[0208] Referring to FIG. **8**, there is a process flow diagram of a method **800** of object location tracking of one object **99** from among many objects **99** in a same vicinity, according to some embodiments. At step **802** of method **800**, a location tracking device **100, 200, 300** is affixed to each of the one object **99** and the many objects **99**. In some embodiments, the one object **99** is a pallet. In some embodiments, the many objects **99** are pallets.

[0209] In some embodiments, the location tracking devices **100, 200, 300** of the one object **99** and the many objects **99** include the same functionality. That is, the one object **99** and the many objects **99** may each have the same type of location tracking device **100, 200, 300** affixed to them, such as location tracking device **100**, for example. That is, the one object **99** and the many objects **99** may each have the same type of location tracking device **100, 200, 300** affixed to them, such as location tracking device **200**, for example. That is, the one object **99** and the many objects **99** may each have the same type of location tracking device **100, 200, 300** affixed to them, such as location tracking device **300**, for example.

[0210] In some embodiments, the location tracking devices **100, 200, 300** of the one object **99** and the many objects **99** include different functionality according to the functionality of a variety of ones of the location tracking devices **100, 200, 300**. That is, the one object **99** and the many objects **99** may carry different types of respective ones of location tracking devices **100, 200, 300** affixed to them, such as location tracking device **100** and location tracking device **200**, for example. The one object **99** and the many objects **99** may respectively carry a combination of any one of location tracking device **100**, location tracking device **200**, and location tracking device **300**, affixed to them, for example.

[0211] The location tracking device **100, 200, 300** is configured to, as described above, detect the movement state of the location tracking device **100, 200, 300**. The location tracking device **100, 200, 300** is further configured to, as described above, in response to detecting the movement state, attempt to establish a connection to identify itself to the external server **155**. The location tracking device **100, 200, 300** is further configured to, as described above, if the connection is established, allow connection with one or more other location tracking devices **100, 200, 300**, so that the one or more other location tracking devices **100, 200, 300** can be identified to the external server **155**.

[0212] In some embodiments, the location tracking device **100, 200, 300** is further configured to, as described above, if the connection is not established, disable all functionality of the location tracking device **100, 200, 300** not required to detect the movement state of the location tracking device **100, 200, 300**.

[0213] In some embodiments, method **800** further includes step **804**. At step **804**, the external server **155** receives device identification data from the location tracking device **100, 200, 300** of the one object **99**. The received device identification data uniquely identifies the one object **99**. In some embodiments, the external server **155** receives device identification data from the location tracking device **100, 200, 300** of the one object **99** and the location tracking devices **100, 200, 300** of the many objects **99**.

[0214] In some embodiments, the device identification data of the location tracking devices **100, 200, 300** of the many objects **99** is received from the location tracking device **100, 200, 300** of the one object **99**.

[0215] In some embodiments, method **800** further includes step **806**. At step **806**, the external server **155** is accessed to identify a location of the location tracking device **100, 200, 300** of the one object **99**. Identifying the location of the location tracking device **100, 200, 300** of the one object **99** identifies a location of the one object **99**. In some embodiments, the external server **155** is accessed to identify a location of the location tracking devices **100, 200, 300** of the many objects **99**. Identifying the location of the location tracking devices **100, 200, 300** of the many objects **99** identifies a location of the many objects **99**.

[0216] Referring back to FIG. **1**, some embodiments relate to a system **16** for object **99** location tracking, including at least two of any of location tracking devices **100, 200, 300** that are in communication with each other or at least have been in communication with each, where one of the location tracking devices **100, 200, 300** provides or has provided help in the help mode to another one of location tracking devices **100, 200, 300**. Further, some embodiments relate to a system for object **99** location tracking in which system **16** includes the external server **155**. For example, the external server **155** may receive device data from at least one of the location tracking devices **100,**

200, 300 of system **16** (e.g. via cellular data communications or via network **150**), and optionally also transmits device data (such as configuration data) to at least one of the location tracking devices **100, 200, 300** of system **16** (e.g. via cellular data communications or via network **150**). In such embodiments, the server **155** may be in communication with network **150** and at least one of the location tracking devices **100, 200, 300** of system **16** can communicate with the network **150**, so that the external server **155** and at least one of the location tracking devices **100, 200, 300** can communicate with each other when possible. Each of the at least two location tracking devices **100, 200, 300** is carried by an object **99**, which allows the external server **155** to track the location of the object **99**.

[0217] Each of the at least two location tracking devices **100, 200, 300** of system **16** is configured to, as described above, detect the movement state of the location tracking device **100, 200, 300**. Each of the at least two location tracking devices **100, 200, 300** is further configured to, as described above, in response to detecting the movement state, establish (or at least attempt to establish) a connection to identify itself to the external server **155**. Each of the at least two location tracking devices **100, 200, 300** is further configured to, as described above, if the connection is established, allow connection with one or more other location tracking devices **100, 200, 300** of the at least two location tracking devices **100, 200, 300**, so that the one or more other location tracking devices **100, 200, 300** can be identified to the external server **155**.

[0218] The external server **155** is configured to identify a location of each object **99** based on a location of their respective location tracking device **100, 200, 300** of system **16**.

[0219] In some embodiments, the at least two location tracking devices **100, 200, 300** of system **16** include the same functionality. That is, system **16** may include at least two location tracking devices **100**, for example. That is, system **16** may include at least two location tracking devices **200**, for example. That is, system **16** may include at least two location tracking devices **300**, for example.

[0220] In some embodiments, the at least two location tracking devices **100, 200, 300** of system **16** include different functionality. That is, system **16** may include a combination of at least two of any one of location tracking device **100**, location tracking device **200**, and/or location tracking device **300**, for example.

[0221] Numerous variations and/or modifications may be made to the above-described embodiments, without departing from the broad general scope of the present disclosure. The present embodiments are, therefore, to be considered in all respects as illustrative and not restrictive.

Claims

1. A method of pallet location tracking of one pallet from among many pallets in a same vicinity, the method including: affixing a location tracking device to each of the one pallet and the many pallets, the location tracking device being configured to: detect a movement state of the location tracking device; in response to detecting the movement state, attempt to establish a connection to identify itself to an external server; and if the connection is established, then allow connection with one or more other location tracking devices, so that the one or more other location tracking devices can be identified to the external server.
2. The method of claim 1, wherein the location tracking device is further configured to, if the connection is not established, disable all functionality of the location tracking device not required to detect the movement state of the location tracking device.
3. The method of claim 1, further including, after the affixing, receiving at the external server device identification data from the location tracking device of the one pallet, wherein the device identification data uniquely identifies the one pallet.
4. The method of claim 1, further including, after the affixing, receiving at the external server

device identification data from the location tracking device of the one pallet and the location tracking devices of the many pallets.

5. The method of claim 4, wherein the device identification data from the location tracking devices of the many pallets is received from the location tracking device of the one pallet.

6. The method of claim 3, further including, after the receiving, accessing the external server to identify a location of the one pallet.

7. The method of claim 1, wherein the movement state is detected based on output of a movement sensor in each location tracking device, wherein detection of the movement state includes detecting when the location tracking device has moved for a first minimum time period followed by no movement for a second minimum time period.

8. The method of claim 7, wherein the first minimum time period is between about 5 seconds to about 30 seconds.

9. The method of claim 7, wherein the second minimum time period is between about 5 seconds to about 30 seconds or between about 5 seconds to about 3 minutes.

10. A system for pallet location tracking including: multiple location tracking devices, each location tracking device carried by a respective pallet, each location tracking device being configured to: detect a movement state of the respective location tracking device; in response to detecting the movement state, attempt to establish a connection to identify itself to an external server; and if the connection is established, then allow connection with one or more other location tracking devices, so that the one or more other location tracking devices can be identified to the external server; and the external server, wherein the external server is configured to identify a location of each pallet based on a location of the respective identified location tracking device carried by the pallet.

11. The system of claim 10, wherein each location tracking device includes a movement sensor and the movement state is detected based on output of the movement sensor, wherein detection of the movement state includes detecting when the location tracking device has moved for a first minimum time period followed by no movement for a second minimum time period.

12. The system of claim 11, wherein the first minimum time period is between about 5 seconds to about 30 seconds.

13. The system of claim 11, wherein the second minimum time period is between about 5 seconds to about 30 seconds or between about 5 seconds to about 3 minutes.

14. The system of claim 10, wherein the location tracking device is further configured to, if the connection is not established, disable all functionality of the location tracking device not required to detect the movement state of the location tracking device.

15. The system of claim 10, wherein the multiple location tracking devices are configured to cooperate to establish a network of wirelessly connected location tracking devices.

16. A system for object location tracking including: multiple location tracking devices, each location tracking device carried by a respective object, each location tracking device being configured to: detect a movement state of the respective location tracking device; in response to detecting the movement state, attempt to establish a connection to identify itself to an external server; and if the connection is established, then allow connection with one or more other location tracking devices, so that the one or more other location tracking devices can be identified to the external server; and the external server, wherein the external server is configured to identify a location of each object based on a location of the respective identified location tracking device carried by the object.

17. The system of claim 16, wherein the location tracking device is further configured to, if the connection is not established, disable all functionality of the location tracking device not required to detect the movement state of the location tracking device.

18. The system of claim 16, wherein each location tracking device includes a movement sensor and the movement state is detected based on output of the movement sensor, wherein detection of the

movement state includes detecting when the location tracking device has moved for a first minimum time period followed by no movement for a second minimum time period, wherein the first minimum time period is between about 5 seconds to about 30 seconds, and wherein the second minimum time period is between about 5 seconds to about 30 seconds or between about 5 seconds to about 3 minutes.

19. The system of claim 16, wherein each location tracking device is configured to attempt to establish the connection with a local wireless communication device that is connected to a network to communicate with the external server; and wherein fewer than all of the multiple location tracking devices are further configured to communicate with the external server without connecting to the local wireless communication device.

20. The system of claim 16, wherein the multiple location tracking devices include different types of location tracking devices that include different communication functionality.
